

MTH 2215- Midterm Exam

NAME: [REDACTED]

TROY ID NUMBER: [REDACTED]

SIGNATURE: [REDACTED]

General instructions:

- Shows all your work to receive full credit.
- No discussion is allowed in the exam room.

PROBLEM	POINTS	SCORE
1	40	40
2	40	40
3	40	40
4	40	40
5	40	40
6	30	30
7	40	40
8	30	30
TOTAL	300	300

1. (40 points) i) Find the truth table of the proposition

$$\neg(p \wedge q) \vee (\neg q \vee r)$$

ii) Determine the value of each of the following proposition given:

$$P(x, y) : (x < y) \rightarrow (x^2 < y^2), x, y \text{ are real numbers}$$

a) $\forall x \forall y P(x, y)$

b) $\exists y \forall x P(x, y)$

i) The truth table of the proposition. $\neg(p \wedge q) \vee (\neg q \vee r)$.

P	q	r	$p \wedge q$	$\neg(p \wedge q)$	$q \vee r$	$\neg q \vee r$	Result
T	T	T	T	F	///	T	T
T	T	F	T	F	///	F	F ✓
T	F	T	F	T	/	T	T
T	F	F	F	T	/	T	T
F	T	T	F	T	/	T	T
F	T	F	F	T	/	F	T
F	F	T	F	T	/	T	T
F	F	F	F	T	///	T	T

ii, $P(x, y) : (x < y) \rightarrow (x^2 < y^2), x, y \in \mathbb{R}$

a, $\forall x \forall y P(x, y)$.

For every real x, y , if $x < y$, then $x^2 < y^2$.

Take a counterexample: $x = -2; y = 1$.
 $\rightarrow -2 < 1$ (true).
 But $(-2)^2 = 4; 1^2 = 1$.
 $\Rightarrow 4 < 1$ (false) ✓

Hence $P(-2, 1)$ is false.

Therefore: $\forall x \forall y P(x, y)$ is false.

b, $\exists y \forall x P(x, y)$

There exists a real number q such that for every real x
 $(x < q) \rightarrow (x^c < q^c)$.

We show this is impossible.

For any real q , choose:

• if $q \geq 0$, let $x = q - 1$.

• if $q < 0$, let $x = q - 1$.

In both case: $x < q$ and $x^c < q^c$.

$\Rightarrow$ The implication fails.

Thus no such q exists.

Therefore: $\exists q \forall x P(x, q)$ is false. ✓

2. (40 points) i) Prove the following set identity:

$$(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

ii) Is the set identity $(A - B) \cup (B - C) = (A - C)$ true? If it is true give a proof, otherwise give a counterexample to show that it is false.

$$i., (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

$$B - A = \{x \mid x \in B \text{ and } x \notin A\}$$

So, $(A - B) \cup (B - A)$ contains all elements that belong to exactly one of the two sets A or B .

The right-hand side: $(A \cup B) - (A \cap B)$.

$\Rightarrow \{x \in A \cup B : x \text{ is in at least one of } A, B\}$
 $\{x \notin A \cap B : x \text{ is not in both}\}$

$\Rightarrow x$ belongs to exactly one of A or B .

Thus both sides describe the same sets:

$$(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

ii, Answer: false:

$$\text{Let } A = \{1\}; B = \{2\}; C = \{3\}.$$

$$\text{Left-hand side: } (A - B) \cup (B - C) = (\{1\} - \{2\}) \cup (\{2\} - \{3\}) = \{1\} \cup \{2\} = \{1, 2\}$$

$$\text{Right-hand side: } (A - C) = \{1\} - \{3\} = \{1\}$$

Since $\{1, 2\} \neq \{1\}$, the identity is false. ✓

3. (40 points) i) For each of the following predicates, find all the values in the domain $D = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ such that the predicate is true:

a) $Q(n): (n^2 < 15) \rightarrow (n \text{ is an even number})$

b) $S(n): (2n > 10) \vee (n|8)$

ii) Prove that $n = \left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil$ for any positive integer n .

i) a, $Q(n): (n^2 < 15) \rightarrow n \text{ is an even number}$.

- Test n where $n^2 < 15$ is true. These are $n \in \{1, 2, 3\}$.

+ $n = 1: (1 < 15) \rightarrow (1 \text{ is even}) = T \rightarrow F = \text{False}$

+ $n = 2: (4 < 15) \rightarrow (2 \text{ is even}) = T \rightarrow T = \text{True}$

+ $n = 3: (9 < 15) \rightarrow (3 \text{ is even}) = T \rightarrow F = \text{False}$

- Test n where $n^2 < 15$ is false: These are $n \in \{4, 5, 6, 7, 8, 9\}$.

+ Since the hypothesis is false, the implication is true.

$\Rightarrow$ Values for which $Q(n)$ is true: $\{2, 4, 5, 6, 7, 8, 9\}$ ✓

b, $S(n): (2n > 10) \vee (n|8)$.

- Check $2n > 10$. This is true for $n > 5$, so $n \in \{6, 7, 8, 9\}$.

- Check $n|8$. The divisors of 8 in the domain are $n \in \{1, 2, 4, 8\}$.

- Union of sets: $\{6, 7, 8, 9\} \cup \{1, 2, 4, 8\} = \{1, 2, 4, 6, 7, 8, 9\}$.

$\Rightarrow$ Values for which $S(n)$ is true: $\{1, 2, 4, 6, 7, 8, 9\}$ ✓

ii, $n = \left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil$

Case 1: n is even.

Let $n = 2k$ for some integer k .

+ $\left\lfloor \frac{2k}{2} \right\rfloor = \left\lfloor k \right\rfloor = k$

+ $\left\lceil \frac{2k}{2} \right\rceil = \left\lceil k \right\rceil = k$

+ Sum: $k + k = 2k = n$ (correct)

Case 2: n is odd.

Let $n = 2k + 1$ for some integer k .

+ $\left\lfloor \frac{2k+1}{2} \right\rfloor = \left\lfloor k + 0.5 \right\rfloor = k$

+ $\left\lceil \frac{2k+1}{2} \right\rceil = \left\lceil k + 0.5 \right\rceil = k + 1$

+ Sum: $k + (k + 1) = 2k + 1 = n$ (correct)

$\Rightarrow$ Since the identity holds for both even and odd integers, the proof is complete

4. (40 points) Which of the following functions are injective? surjective? bijective? Explain your answers.

$$- f: \mathbb{N} \rightarrow \mathbb{N}, f(x) = \begin{cases} \frac{n+1}{2}, & n \equiv 1 \pmod{2} \\ \frac{n}{2}, & n \equiv 0 \pmod{2} \end{cases} \quad \text{where } \mathbb{N} = \{0, 1, 2, 3, \dots\}$$

$$- g: \mathbb{R} \rightarrow \mathbb{Z}, g(x) = \lfloor x+1 \rfloor$$

$$- h: \mathbb{Z}^+ \rightarrow \mathbb{Z}, h(x) = x^2 - 1 \quad \text{where } \mathbb{Z}^+ = \{1, 2, 3, \dots\}$$

$$\textcircled{*} f: \mathbb{N} \rightarrow \mathbb{N}, f(x) = \begin{cases} \frac{n+1}{2}, & n \equiv 1 \pmod{2} \\ \frac{n}{2}, & n \equiv 0 \pmod{2} \end{cases} \quad \text{where } \mathbb{N} = \{0, 1, 2, 3, \dots\}$$

- Injective : No. $\checkmark$
 + Example: $f(0) = \frac{0}{2} = 0$ and $f(-1)$ isn't in the domain, but check

$$f(1) \text{ and } f(2).$$

$$+ f(1) = \frac{1+1}{2} = 1.$$

$$+ f(2) = \frac{2}{2} = 1.$$

+ Since $f(1) = f(2)$ but $1 \neq 2 \Rightarrow$ it's not injective.

- Surjective : Yes. $\checkmark$
 + For any $y \in \mathbb{N}$, we can choose $n = 2y$. Since n is even, $f(\frac{2y}{2}) = \frac{2y}{2} = y$. Every element in the codomain is hit.

- Bijective : No (it's not injective).

$$\textcircled{*} g: \mathbb{R} \rightarrow \mathbb{Z}, g(x) = \lfloor x+1 \rfloor$$

- Injective : No. $\checkmark$

+ The floor function maps many real numbers to the same integer

$$+ \text{Example: } g(0, 1) = \lfloor 1, 1 \rfloor = 1 \text{ and } g(0, 2) = \lfloor 1, 2 \rfloor = 1.$$

- Surjective : Yes.
 + For any integer $z \in \mathbb{Z}$, let $x = z - 1$. Then $g(z-1) = \lfloor (z-1)+1 \rfloor = \lfloor z \rfloor = z$.

- Bijective : No (it's not injective). $\checkmark$

④ $h: \mathbb{Z}^+ \rightarrow \mathbb{Z}, h(x) = x^2 - 1$ where $\mathbb{Z}^+ = \{1; 2; 3; \dots\}$.

→ Injective: Yes:

+ Suppose $h(a) = h(b) \Rightarrow a^2 - 1 = b^2 - 1 \Rightarrow a^2 = b^2$.

+ Since the domain is restricted to positive integers ($a, b > 0$) then $a = b$.

→ Surjective: No.

+ The codomain is all integers $\mathbb{Z}$, but $x^2 - 1$ cannot produce all integers.

+ Example: There is no $x \in \mathbb{Z}^+$ such that $x^2 - 1 = 1$ (because $x^2 = 2$ has no integer solution)

+ Bijective: No (it's not surjective).
a $x^2 - 1 = -5$

5. (40 points) What is the least integer n such that $f(x)$ is $O(x^n)$ for each of the following functions? Explain in detail.

a) $f(x) = x^{5/2} + x^2 + 5$

b) $f(x) = 2x^5 + x^4 \ln x - 2x^2$

a, $f(x) = x^{5/2} + x^2 + 5$.

+ The dominant term. As x becomes very large, x^2 grows faster than $x^{1.5}$ and 5.

+ Big-O bound: Since x^2 is the highest power, $f(x)$ is $O(x^2)$.

+ Conclusion: The least integer n is 2. ✓

b, $f(x) = 2x^5 + x^4 \ln x - 2x^2$.

+ We know that every $\epsilon > 0$, $\ln x$ grows slower than x^ϵ .

+ Therefore, $x^4 \ln x$ grows faster than x^4 , but slower than $x^{4+\epsilon}$.

+ If we pick $n=4$, then $\frac{x^4 \ln x}{x^4} = \ln x$, which approaches ∞ as $x \rightarrow \infty$.

Thus, $f(x)$ is not $O(x^4)$.

+ If we pick $n=5$, then $\frac{x^4 \ln x}{x^5} = \frac{\ln x}{x}$, which approaches 0 as $x \rightarrow \infty$.

Thus $f(x)$ is $O(x^5)$.

+ Conclusion: The least integer n is 5. ✓

6. (30 points) Let x, y be positive real numbers such that $x^2 - y^2 = 1$. Prove that x or y (or both) is not an integer.

We have: $x^2 - y^2 = (x-y)(x+y) = 1$.

Assume, for contradiction, that both x and y are integers.

Since, $x, y \in \mathbb{Z}$, it follows that x^2 and y^2 are integers.

Hence: $x^2 - y^2 = 1$ implies: $x^2 = y^2 + 1$.

Thus two perfect squares differ by 1.

Now let $y = n$, where $n \in \mathbb{Z}^+$. Then $x^2 = n^2 + 1$.

Since: $(n+1)^2 = n^2 + 2n + 1 > n^2 + 1$ (because $n \geq 1$).

We obtain: $n^2 < x^2 < (n+1)^2$

So $x^2 = n^2 + 1$ lies strictly between two consecutive perfect squares n^2 and $(n+1)^2$, which is impossible. Therefore, our

assumption that both x and y are integers is false.

Hence at least one of x or y is not an integer.

$\Rightarrow$ Therefore, x or y (or both) is not an integer. $\checkmark$

7. (40 points) i) Show the steps of using Euclidean algorithm to find $\gcd(1411, 2703)$
 ii) Find the binary representation of $(8FB)_{16}$
 iii) Find the octal representation of $(10010110101)_2$

i, $\gcd(1411, 2703)$.

Apply the Euclidean algorithm:

$$\begin{aligned} 2703 &= 1411 \cdot 1 + 1292 \\ 1411 &= 1292 \cdot 1 + 119 \\ 1292 &= 119 \cdot 10 + 102 \\ 119 &= 102 \cdot 1 + 17 \\ 102 &= 17 \cdot 6 + 0 \end{aligned}$$

$\Rightarrow \gcd(1411, 2703) = 17 \checkmark$

ii, Convert: $8 = 1000$, $F = 1111$, $B = 1011$.

Thus, $(8FB)_{16} = (1000 \ 1111 \ 1011)_2$

Therefore, $(8FB)_{16} = (100011111011)_2 \checkmark$

iii, $(10010110101)_2 = 010 \ 010 \ 110 \ 101$.

Convert: $010 = 2$, $010 = 2$, $110 = 6$, $101 = 5$.

Hence: $(10010110101)_2 = (2265)_8$.

Therefore: $(10010110101)_2 = (2265)_8 \checkmark$

8. (30 points) Show the steps of using modular exponentiation algorithm to find $13^{61} \bmod 65$

Step 1: Convert the exponent to binary.

The exponent is 61. We convert it to binary by finding the powers of 2. $61 = 32 + 16 + 8 + 4 + 1$.

$$61 = (111101)_2$$

Step 2: Calculate the square of the base modulo 65.

We compute $13^{2^k} \pmod{65}$ for each bit position:

$$+ 13^1 = 13 \pmod{65}$$

$$+ 13^2 = 169 = 59 \pmod{65}$$

$$+ 13^4 = 39^2 = 1521 = 26 \pmod{65}$$

$$+ 13^8 = 26^2 = 676 = 26 \pmod{65}$$

$$+ 13^{16} = 26^2 = 26 \pmod{65}$$

$$+ 13^{32} = 26^2 = 26 \pmod{65}$$

Step 3: Multiply the results according to the binary expansion.

We multiply the values corresponding to the "1" bits in the binary representation $(111101)_2$.

$$13^{61} = 13^{32} \cdot 13^{16} \cdot 13^8 \cdot 13^4 \cdot 13^1 \pmod{65}$$

Substitute in the values calculated in step 2:

$$1, 13^2 \cdot 13^{16} = 26 \cdot 26 = 26 \pmod{65}$$

$$2, 26 \cdot 13^8 = 26 \cdot 26 = 26 \pmod{65}$$

$$3, 26 \cdot 13^4 = 26 \cdot 26 = 26 \pmod{65}$$

$$4, 26 \cdot 13^1 = 26 \cdot 13 = 338 \pmod{65}$$

Step 4: Final calculation.

$$338 / 65 = 5 \text{ remainder } 13$$

$$338 = 5 \cdot 65 + 13$$

$$\text{Answer: } 13^{61} \equiv 13 \pmod{65} \quad \checkmark$$